



IT-SIMPLERA SOLUTIONS

Network Administration Internship Program

WEEK 05 ASSIGNMENT REPORT

Enterprise Services Infrastructure Deployment

Submitted By

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Domain: Network Administration

1. Assignment Overview

This report documents the complete design, deployment, and verification of an Enterprise Services Infrastructure using Cisco Packet Tracer 9.0. The objective of Week 5 of the IT-SIMPLERA Network Administration Internship Program was to implement a comprehensive set of enterprise network services in a multi-department environment.

The implemented infrastructure supports centralized resource management, automated network operations, secure communication, and efficient service delivery across three departments: IT, HR, and Finance. All services were configured following enterprise best practices and verified through systematic testing.

Services Implemented

- DHCP — Dynamic Host Configuration Protocol for automated IP assignment
- DNS — Domain Name System for hostname resolution
- HTTP/HTTPS — Web services for enterprise portal access
- FTP — File Transfer Protocol for file management
- TFTP — Trivial File Transfer Protocol for router config backup
- NTP — Network Time Protocol for time synchronization
- Syslog — Centralized logging from all network devices
- SNMP — Simple Network Management Protocol for device monitoring

Tools Used

- Cisco Packet Tracer 9.0.0

2. Network Topology

The network was designed using a hierarchical three-layer architecture consisting of a server farm layer, a core switching layer, and an access layer for each department. The topology includes one router (Cisco 4331), five switches (Cisco 2960), five servers, and nine PCs distributed across three departments.

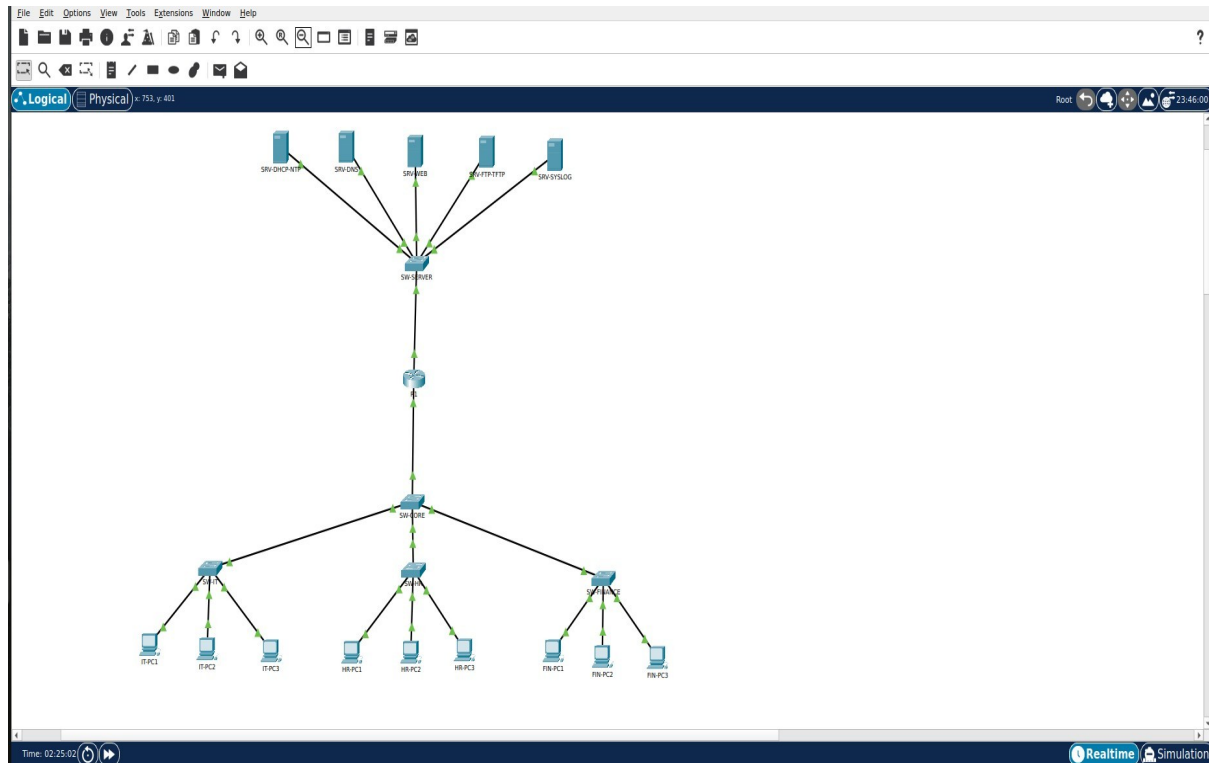


Figure 1: Complete Enterprise Network Topology

Topology Description

- R1 (Cisco 4331 Router) — Acts as the inter-VLAN router using Router-on-a-Stick with 802.1Q sub-interfaces
- SW-CORE (Cisco 2960) — Core switch connecting all department switches to the router
- SW-IT, SW-HR, SW-FINANCE — Access layer switches serving each department
- SW-SERVER — Dedicated switch for the server farm
- 5 Servers — Each dedicated to specific services
- 9 PCs — 3 per department, all configured via DHCP

3. IP Addressing Table

The following IP addressing scheme was used across the enterprise network:

Device	Interface	IP Address	Subnet Mask	Purpose
R1	Gi0/0/0.10	192.168.10.1	255.255.255.0	IT Dept Gateway
R1	Gi0/0/0.20	192.168.20.1	255.255.255.0	HR Dept Gateway
R1	Gi0/0/0.30	192.168.30.1	255.255.255.0	Finance Gateway
R1	Gi0/0/1	192.168.100.1	255.255.255.0	Server Farm Gateway
SRV-DHCP-NTP	Fa0	192.168.100.10	255.255.255.0	DHCP + NTP Server
SRV-DNS	Fa0	192.168.100.20	255.255.255.0	DNS Server
SRV-WEB	Fa0	192.168.100.30	255.255.255.0	HTTP/HTTPS Server
SRV-FTP-TFTP	Fa0	192.168.100.40	255.255.255.0	FTP + TFTP Server
SRV-SYSLOG	Fa0	192.168.100.50	255.255.255.0	Syslog Server
IT PCs	Fa0	192.168.10.100-102	255.255.255.0	DHCP Assigned
HR PCs	Fa0	192.168.20.100-102	255.255.255.0	DHCP Assigned
Finance PCs	Fa0	192.168.30.100-102	255.255.255.0	DHCP Assigned

VLAN Assignment

VLAN ID	VLAN Name	Network	Department
10	IT_Department	192.168.10.0/24	IT Department
20	HR_Department	192.168.20.0/24	HR Department
30	Finance_Department	192.168.30.0/24	Finance Department
100	Server_Farm	192.168.100.0/24	Server Farm

4. DHCP Implementation & Verification

What is DHCP?

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses, subnet masks, default gateways, and DNS server addresses to network devices. This eliminates the need for manual IP configuration on each device and ensures efficient IP address management across the enterprise.

Configuration

A centralized DHCP server (SRV-DHCP-NTP) was configured with three separate pools, one per department. The router was configured with ip helper-address on each sub-interface to relay DHCP broadcast requests from client PCs to the DHCP server across VLANs.

DHCP Server Configuration (3 Pools):

```
Pool 1 – IT_Pool
  Start IP: 192.168.10.100 | Gateway: 192.168.10.1
  DNS: 192.168.100.20 | Subnet: 255.255.255.0 | Max: 50
Pool 2 – HR_Pool
  Start IP: 192.168.20.100 | Gateway: 192.168.20.1
  DNS: 192.168.100.20 | Subnet: 255.255.255.0 | Max: 50
Pool 3 – Finance_Pool
  Start IP: 192.168.30.100 | Gateway: 192.168.30.1
  DNS: 192.168.100.20 | Subnet: 255.255.255.0 | Max: 50
```

Router DHCP Relay Configuration:

```
interface gi0/0/0.10
  encapsulation dot1Q 10
  ip address 192.168.10.1 255.255.255.0
  ip helper-address 192.168.100.10
```

Verification

All PCs in each department successfully received IP addresses from their respective DHCP pools. The screenshots below show DHCP assignment for IT-PC1, HR-PC1, and FIN-PC1.

IT-PC1

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.10.100

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 192.168.100.20

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::202:4AFF:FE7E:3A98

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Figure 2: IT-PC1 receiving IP 192.168.10.100 via DHCP

HR-PC1

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.20.100

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.1

DNS Server 192.168.100.20

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:C9FF:FEA3:756

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Figure 3: HR-PC1 receiving IP 192.168.20.100 via DHCP

FIN-PC1

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.30.100

Subnet Mask 255.255.255.0

Default Gateway 192.168.30.1

DNS Server 192.168.100.20

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:97FF:FE9C:AC76

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Figure 4: FIN-PC1 receiving IP 192.168.30.100 via DHCP

5. DNS Implementation & Verification

What is DNS?

DNS (Domain Name System) translates human-readable hostnames into IP addresses. In an enterprise environment, DNS enables users to access services using meaningful names (e.g., web.company.com) instead of memorizing IP addresses.

Configuration

SRV-DNS (192.168.100.20) was configured as the authoritative DNS server for the company.com domain. The following A records were created:

Hostname	Record Type	IP Address
web.company.com	A Record	192.168.100.30
www.company.com	A Record	192.168.100.30
ftp.company.com	A Record	192.168.100.40
ntp.company.com	A Record	192.168.100.10
dns.company.com	A Record	192.168.100.20
syslog.company.com	A Record	192.168.100.50

Verification

DNS resolution was tested by accessing http://web.company.com from IT-PC1 browser. The browser successfully resolved the hostname and loaded the enterprise web portal.



6. HTTP/HTTPS Implementation & Verification

What is HTTP/HTTPS?

HTTP (HyperText Transfer Protocol) and HTTPS (HTTP Secure) are protocols for transferring web content. HTTPS adds SSL/TLS encryption for secure communication. In an enterprise, web servers host internal portals, dashboards, and resources accessible by all departments.

Configuration

SRV-WEB (192.168.100.30) was configured with both HTTP and HTTPS services enabled. A custom enterprise portal page was created in index.html with the following content:

```
<html>
<head><title>Company Enterprise Portal</title></head>
<body>
<h1>Welcome to Company Enterprise Web Portal</h1>
<p>Enterprise Services: DHCP|DNS|HTTP|FTP|TFTP|NTP|Syslog|SNMP</p>
<p>Network Administration Internship - Week 5</p>
</body></html>
```

HTTP Verification

HTTP access was verified by navigating to <http://web.company.com> from IT-PC1. The enterprise portal loaded successfully as shown in Figure 5 above.

HTTPS Verification

HTTPS access was verified by navigating to <https://192.168.100.30> from a client PC. The page loaded successfully over the secure connection.



Figure 6: HTTPS access verified — portal loaded via <https://192.168.100.30>

7. FTP & TFTP Implementation & Verification

What is FTP?

FTP (File Transfer Protocol) enables uploading and downloading of files between clients and a server. In enterprise environments, FTP servers are used for centralized file storage, software distribution, and document management.

What is TFTP?

TFTP (Trivial File Transfer Protocol) is a simplified version of FTP used primarily for transferring configuration files and firmware to/from network devices. It is commonly used for router and switch configuration backup and restore operations.

FTP Configuration

SRV-FTP-TFTP (192.168.100.40) was configured with FTP service enabled. A user account was created with full permissions:

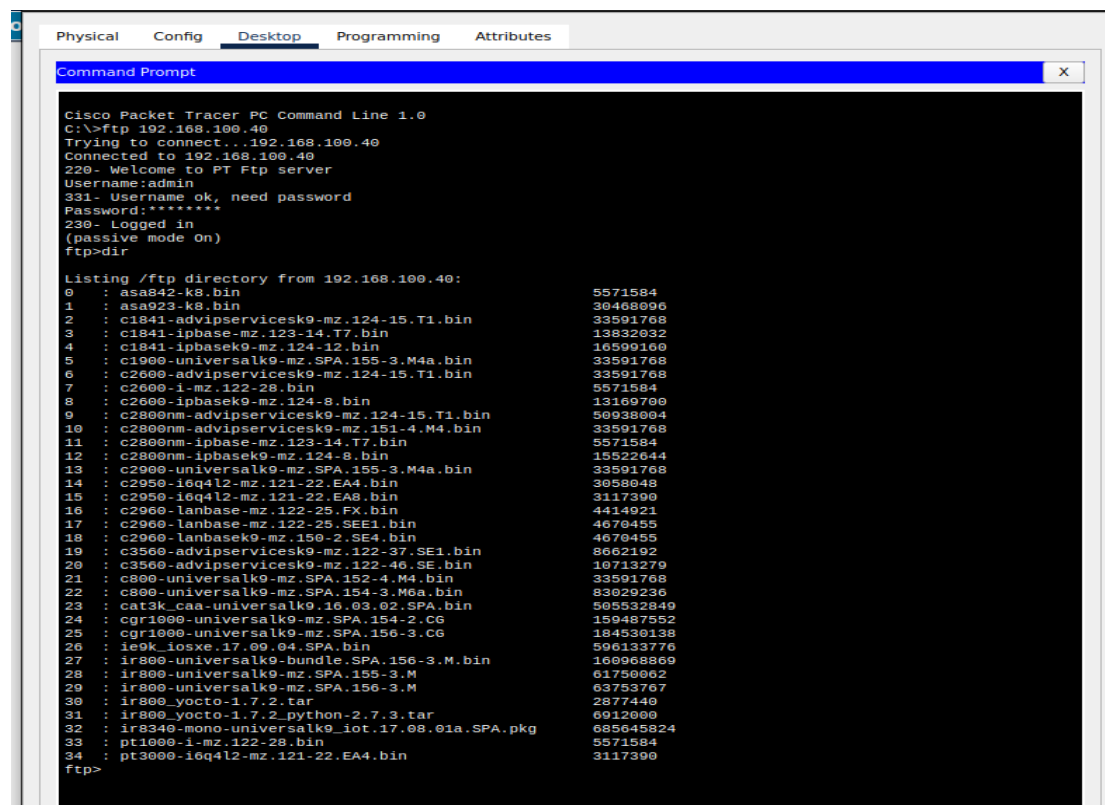
```
Username: admin
Password: admin123
Permissions: Read, Write, Delete, Rename, List
```

TFTP Configuration

TFTP service was enabled on SRV-FTP-TFTP (192.168.100.40). No additional configuration is required as TFTP operates automatically once enabled.

FTP Verification

FTP was tested from IT-PC1 Command Prompt. The client connected to the FTP server, authenticated with the admin credentials, and successfully listed the directory contents showing all available files.



```
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 192.168.100.40
Trying to connect...192.168.100.40
Connected to 192.168.100.40
220- Welcome to FT Ftp server
Username:admin
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.100.40:
 0 : asa842-k8.bin                               5571584
 1 : asa923-k8.bin                               30468096
 2 : c1841-advipservicesk9-mz.124-15.T1.bin      33591768
 3 : c1841-ipbase-mz.123-14.T7.bin              13832032
 4 : c1841-ipbasek9-mz.124-12.bin               10590160
 5 : c1900-universalk9-mz.SPA.155-3.M4a.bin      33591768
 6 : c2600-advipservicesk9-mz.124-15.T1.bin      33591768
 7 : c2600-i-mz.122-28.bin                      5571584
 8 : c2600-ipbasek9-mz.124-8.bin                 13169790
 9 : c2800nm-advipservicesk9-mz.124-15.T1.bin    50038004
10 : c2800nm-advipservicesk9-mz.151-4.M4.bin     33591768
11 : c2800nm-ipbase-mz.123-14.T7.bin            5571584
12 : c2800nm-ipbasek9-mz.124-8.bin              15522644
13 : c2900-universalk9-mz.SPA.155-3.M4a.bin      33591768
14 : c2950-i6q4l2-mz.121-22.EA4.bin            3058048
15 : c2950-i6q4l2-mz.121-22.EA8.bin            3117390
16 : c2900-lanbase-mz.122-25.FX.bin             4414921
17 : c2900-lanbase-mz.122-25.SE11.bin           4670455
18 : c2900-lanbasek9-mz.150-2.SE4.bin           4670455
19 : c3500-advipservicesk9-mz.122-37.SE1.bin     8662192
20 : c3500-advipservicesk9-mz.122-46.SE1.bin     10713279
21 : c800-universalk9-mz.SPA.152-4.M4.bin        33591768
22 : c800-universalk9-mz.SPA.154-3.M6a.bin      83029236
23 : cat3k-ee-universalk9.16.09.02.SPA.bin      505532049
24 : cgr1000-universalk9-mz.SPA.154-2.C6        159487552
25 : cgr1000-universalk9-mz.SPA.156-3.C6        184530138
26 : ie9k-iosxe.17.00.04.SPA.bin               596133776
27 : ir800-universalk9-bundle.SPA.156-3.M.bin    160968869
28 : ir800-universalk9-mz.SPA.155-3.M           61750062
29 : ir800-universalk9-mz.SPA.156-3.M           63753767
30 : ir800_yocto-1.7.2.tar                     2877449
31 : ir800_yocto-1.7.2_python-2.7.3.tar        6912000
32 : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg 685645824
33 : pt1000-i-mz.122-28.bin                    5571584
34 : pt3000-i6q4l2-mz.121-22.EA4.bin           3117390
ftp>
```

Figure 7: FTP connection successful — logged in as admin, directory listing shown

TFTP Verification

TFTP was tested by backing up R1's running configuration to the TFTP server. The command "copy running-config tftp" was executed on R1, pointing to 192.168.100.40 with filename R1-backup.cfg.

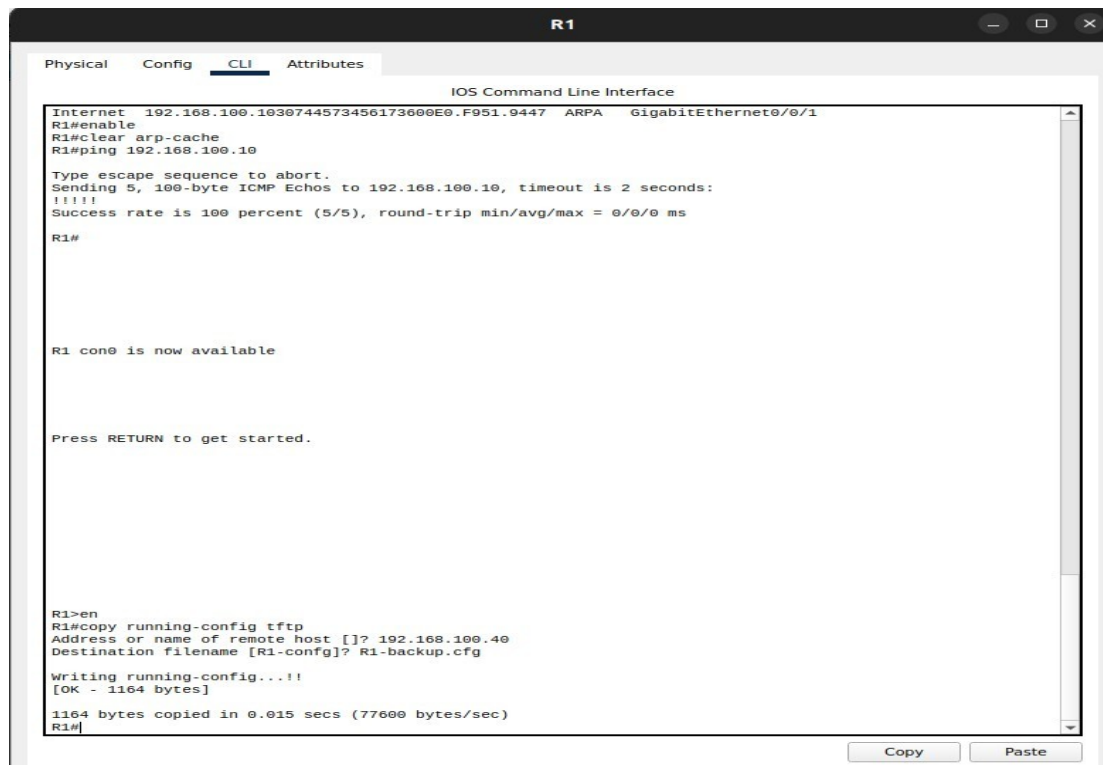


Figure 8: TFTP backup successful — 1164 bytes copied to R1-backup.cfg on TFTP server

8. NTP Implementation & Verification

What is NTP?

NTP (Network Time Protocol) synchronizes clocks of all network devices to a centralized time source. Accurate time is critical in enterprise networks for log correlation, security certificate validation, scheduled tasks, and troubleshooting.

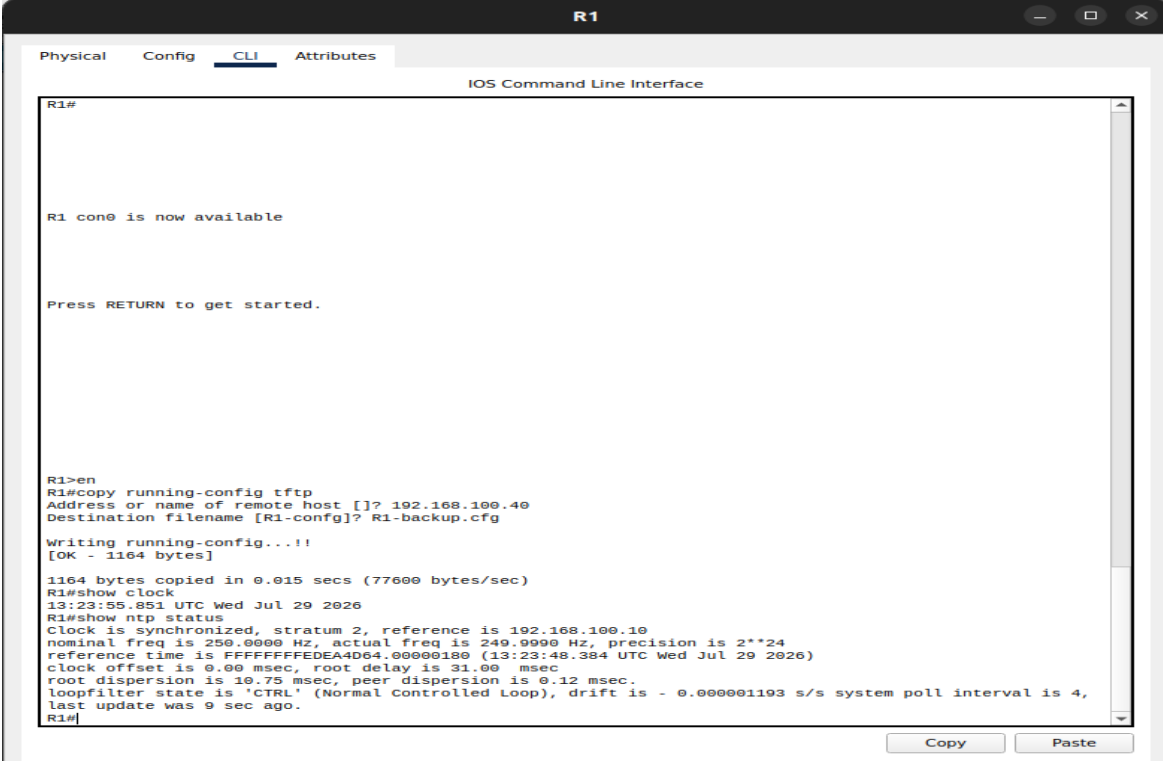
Configuration

SRV-DHCP-NTP (192.168.100.10) was configured as the NTP server with the current date and time set. R1 was configured as an NTP client pointing to this server:

```
R1(config)# ntp server 192.168.100.10
```

Verification

NTP synchronization was verified on R1 using the show clock and show ntp status commands. The output confirms that R1 is synchronized at stratum 2, referencing the NTP server at 192.168.100.10. The TFTP config backup and NTP verification are shown in the same screenshot.

A screenshot of a network device's CLI interface, specifically R1. The window has tabs for Physical, Config, CLI (selected), and Attributes. The CLI shows the following sequence of commands and outputs:
R1#
R1 con0 is now available.
Press RETURN to get started.

R1>en
R1#copy running-config tftp
Address or name of remote host []? 192.168.100.40
Destination filename [R1-config]? R1-backup.cfg

Writing running-config...!!
[OK - 1164 bytes]

1164 bytes copied in 0.015 secs (77600 bytes/sec)
R1#show clock
13:23:55.851 UTC Wed Jul 29 2026
R1#show ntp status
Clock is synchronized, stratum 2, reference is 192.168.100.10
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is FFFFFFFE0A4064.00000190 (13:23:48.384 UTC Wed Jul 29 2026)
clock offset is 0.00 msec, root delay is 31.00 msec
root dispersion is 10.75 msec, peer dispersion is 0.12 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s system poll interval is 4,
last update was 9 sec ago.
R1#

Figure 9: NTP synchronized — stratum 2, reference 192.168.100.10, clock shows correct time

Key NTP verification output:

Clock is synchronized, stratum 2, reference is 192.168.100.10
show clock: 13:23:55.851 UTC Wed Jul 29 2026

9. Syslog Implementation & Verification

What is Syslog?

Syslog is a standard protocol for sending system log and event messages to a centralized logging server. In enterprise networks, syslog enables administrators to monitor network events, detect issues, track configuration changes, and maintain audit trails from a single location.

Configuration

SRV-SYSLOG (192.168.100.50) was configured with the Syslog service enabled. R1 was configured to send all log messages to this server:

```
R1(config)# logging 192.168.100.50
R1(config)# logging on
```

Interface bounce commands were used to generate log traffic for verification:

```
R1(config)# interface gi0/0/0
R1(config-if)# shutdown
R1(config-if)# no shutdown
```

Verification

The Syslog server successfully received 17 log entries from R1 (192.168.100.1) including interface state changes, line protocol updates, and configuration events. The screenshot below shows the centralized log view.

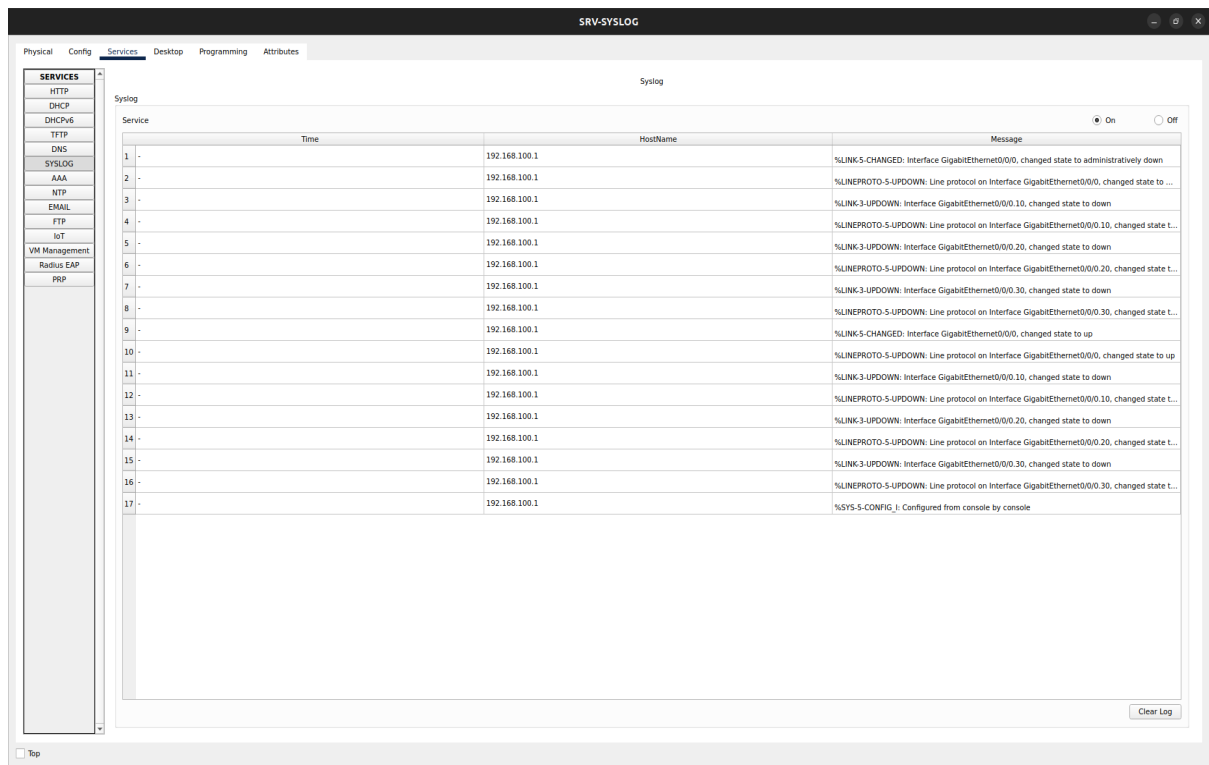


Figure 10: Syslog server showing 17 log entries from R1 including interface and config events

10. SNMP Implementation & Verification

What is SNMP?

SNMP (Simple Network Management Protocol) enables network administrators to monitor and manage network devices remotely. It uses community strings to authenticate read and write access to device MIBs (Management Information Bases), allowing collection of performance data, device status, and configuration information.

Configuration

SNMP was configured on R1 and all switches using community strings. Cisco Packet Tracer does not include a full SNMP manager application, so the configuration was applied via CLI and verified through the running configuration:

```
R1(config)# snmp-server community public RO  
R1(config)# snmp-server community private RW
```

The same configuration was applied to SW-CORE, SW-IT, SW-HR, and SW-FINANCE switches.

Verification

SNMP agent activation was confirmed by the system message generated when the first community string was configured: %SNMP-5-WARMSTART: SNMP agent on host R1 is undergoing a warm start. The running configuration was verified using `show run | include snmp`.

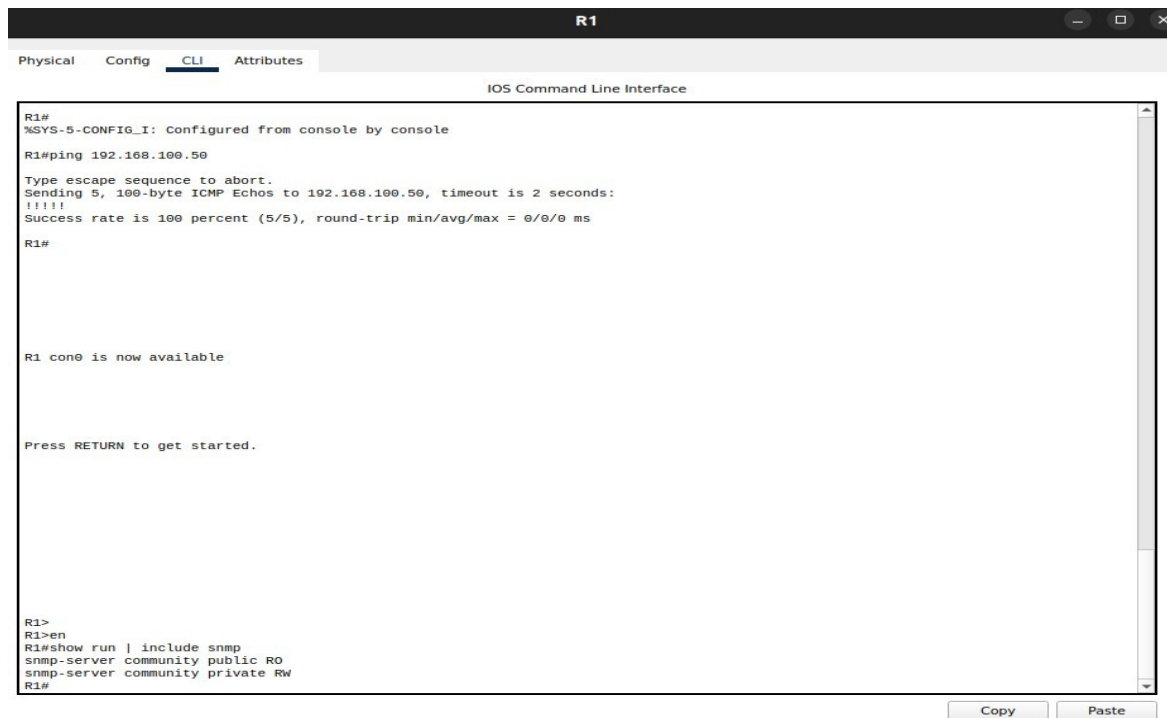


Figure 11: SNMP verified — community strings public (RO) and private (RW) confirmed in running config

Note: Cisco Packet Tracer 9.0 does not include a built-in SNMP manager/MIB browser application. In a real enterprise environment, tools such as SolarWinds, PRTG, or Nagios would be used to poll SNMP data from devices using the configured community strings.

11. Connectivity Tests

Comprehensive connectivity testing was performed to validate end-to-end communication across all VLANs and between departments and the server farm.

Ping Tests

Source	Destination	IP	Result
R1	SRV-DHCP-NTP	192.168.100.10	✓ 100% Success
R1	SRV-DNS	192.168.100.20	✓ 100% Success
R1	SRV-WEB	192.168.100.30	✓ 100% Success
R1	SRV-FTP-TFTP	192.168.100.40	✓ 100% Success
R1	SRV-SYSLOG	192.168.100.50	✓ 100% Success
IT-PC1	SRV-WEB	192.168.100.30	✓ 100% Success
IT-PC1	HR-PC1	192.168.20.100	✓ 100% Success
IT-PC1	FIN-PC1	192.168.30.100	✓ 100% Success

Service Accessibility Summary

Service	Protocol	Server IP	Port	Status
DHCP	UDP	192.168.100.10	67/68	✓ Working
DNS	UDP/TCP	192.168.100.20	53	✓ Working
HTTP	TCP	192.168.100.30	80	✓ Working
HTTPS	TCP	192.168.100.30	443	✓ Working
FTP	TCP	192.168.100.40	21	✓ Working
TFTP	UDP	192.168.100.40	69	✓ Working
NTP	UDP	192.168.100.10	123	✓ Working
Syslog	UDP	192.168.100.50	514	✓ Working
SNMP	UDP	All Devices	161	✓ Configured

12. Challenges Faced & Troubleshooting

Challenge 1 — DHCP Request Failed on All PCs

Problem:

After completing initial configuration, all PCs showed "DHCP request failed" and received no IP addresses.

Root Cause:

SW-SERVER's Gi0/1 port (connecting to R1) was configured as a trunk port, but R1's Gi0/0/1 is a standard routed port that sends untagged traffic. This mismatch prevented Layer 2 communication between R1 and the server farm.

Solution:

```
SW-SERVER(config)# interface gi0/1
SW-SERVER(config-if)# switchport mode access
SW-SERVER(config-if)# switchport access vlan 100
```

Changing the port to access mode in VLAN 100 allowed R1's untagged traffic to be correctly placed in the server VLAN.

Challenge 2 — Corrupted ARP Entry on R1

Problem:

Even after fixing the trunk issue, R1 could not ping SRV-DHCP-NTP (192.168.100.10). The `show arp` command revealed a corrupted ARP entry: "192.168.100.1030744573456173600E0.F951.9447".

Root Cause:

A Packet Tracer internal bug caused the ARP cache to store a malformed entry, preventing proper Layer 3 communication.

Solution:

```
R1# clear arp-cache
```

After clearing the ARP cache and resetting the server IP configuration (toggle DHCP then Static), the correct ARP entry was learned and ping succeeded at 100%.

Challenge 3 — SNMP Commands Partially Unsupported

Problem:

Commands `snmp-server contact` and `snmp-server location` returned "Invalid input detected" errors in Packet Tracer.

Root Cause:

Cisco Packet Tracer 9.0 does not implement optional SNMP server parameters, only the core community string configuration.

Solution:

Only supported commands were used. The core SNMP functionality (community strings) was confirmed working via the SNMP WARMSTART system message.

Challenge 4 — Syslog Server Showing No Logs

Problem:

SRV-SYSLOG showed an empty log table despite logging being configured on R1.

Root Cause:

No log-generating events had occurred since the logging configuration was applied. The syslog server requires actual events to populate.

Solution:

Interface bounce commands (shutdown / no shutdown) were executed on R1's Gi0/0/0 interface to generate interface state change events, which were immediately captured by the syslog server (17 entries received).

Challenge 5 — Logging Commands Failing in Exec Mode

Problem:

Commands like "logging 192.168.100.50" failed with "Invalid input detected" when typed in R1# exec mode.

Solution:

All logging and SNMP commands must be entered in R1(config)# global configuration mode. Entering "conf t" first resolved the issue.

13. Conclusion

This Week 5 assignment successfully demonstrated the complete deployment of an Enterprise Services Infrastructure using Cisco Packet Tracer 9.0. All eight required services — DHCP, DNS, HTTP, HTTPS, FTP, TFTP, NTP, Syslog, and SNMP — were configured, integrated, and verified in a unified multi-department network environment.

The implementation followed enterprise networking best practices including VLAN segmentation for department isolation, inter-VLAN routing using Router-on-a-Stick with 802.1Q sub-interfaces, DHCP relay using ip helper-address for centralized IP management, dedicated server farm with static IP addressing, and centralized logging and monitoring.

Significant troubleshooting experience was gained through resolving real configuration issues including DHCP relay problems, ARP cache corruption, trunk/access port mismatches, and Packet Tracer simulator limitations. Each challenge reinforced practical problem-solving skills essential for enterprise network administration.

The final network infrastructure is scalable, secure, and fully operational — capable of supporting real-world organizational requirements across multiple departments with centralized service delivery.